

Multivariate practical 11

Correspondence analysis

Introduction

Note: unless otherwise stated, calculations are from first principles, i.e. do the matrix algebra manually.

Question One

Here is a contingency table of people's education levels against their preferred movies:

	Action	Drama	Comedy	Documentary
High School	50	60	40	10
Bachelor's	30	80	60	30
Master's	10	40	20	50
Doctorate	1	2	1	4

You can attach it using the following code:

```
```{r}
ratings <- structure(
 c(
 50, 30, 10, 1, 60, 80, 40, 2,
 40, 60, 20, 1, 10, 30, 50, 4),
 dim = c(4L, 4L),
 dimnames = list(
 c("High School", "Bachelor's", "Master's", "Doctorate"),
 c("Action", "Drama", "Comedy", "Documentary"))
)
```
```

1. Calculate the row and column profiles. (1)
2. Compute the χ^2 distance between the row profiles and column profiles, display the results in two separate plots (e.g. in a heat-map style) and comment on the results. (2)
3. Which cell(s) (row and column combinations) indicate the greatest deviance from the assumption of independence of rows and columns? (1)
4. Calculate the principal row and principal column coordinates and display them in a plot. Interpret the results. (4)
5. From the results of `ca(summary(ratings))`, assess the quality of the representation of the data in the first two dimensions. (2)
6. Construct a PCA biplot on the correlation matrix (using any function), and compare it to the results of question four, explaining how and why it differs. (2)
7. Construct the most appropriate CA plot for someone purely interested in the relationship between the different education levels. (1)
8. Under what circumstances would a CA biplot with principal row coordinates and standard column coordinates be difficult to interpret visually, and why? (1)

Question Two

Use `data("HairEyeColor", package = "ca")` to load the `HairEyeColor` dataset.

1. Appropriately transform the data for multiple correspondence analysis. (2)
2. Perform multiple correspondence analysis on the data, using the lambda-adjusted method and interpret the results. (3)